HDOC Day-76

1. Create an enumeration for user roles (ADMIN, USER, GUEST) and display messages based on role.

C Program:

```
#include <stdio.h>
#include <string.h>

enum Role { ADMIN, USER, GUEST };

int main() {
    char input[20];
    enum Role role;

    printf("Enter role: ");
    scanf("%19s", input);

    if (strcmp(input, "ADMIN") == 0)
        role = ADMIN;
    else if (strcmp(input, "USER") == 0)
        role = USER;
    else if (strcmp(input, "GUEST") == 0)
        role = GUEST;
    else {
        printf("Invalid role\n");
        return 0;
    }

    switch (role) {
        case ADMIN: printf("Welcome Admin\n"); break;
        case USER:  printf("Welcome User\n"); break;
        case GUEST: printf("Welcome Guest\n"); break;
    }
    return 0;
}
```

Sample Output:

```
Enter role: GUEST
Welcome Guest
```

2. Print all enum names and integer values of 12 colors using a loop.

C Program:

```
#include <stdio.h>

enum Color {
    RED, BLUE, GREEN, YELLOW, ORANGE, PURPLE,
    PINK, BLACK, WHITE, BROWN, GRAY, CYAN
};

int main() {
    const char *names[] = {
        "RED", "BLUE", "GREEN", "YELLOW", "ORANGE", "PURPLE",
```

```
        "PINK", "BLACK", "WHITE", "BROWN", "GRAY", "CYAN"
    };

    for (enum Color color = RED; color <= CYAN; color++) {
        printf("%s=%d\n", names[color], color);
    }
    return 0;
}
```

Sample Output:

```
RED=0
BLUE=1
GREEN=2
YELLOW=3
ORANGE=4
PURPLE=5
PINK=6
BLACK=7
WHITE=8
BROWN=9
GRAY=10
CYAN=11
```